

Problem scoping

Audio/Video Transcription to Mom/ Lecture Notes/ Interview Q&A

Use case:- predicting the lecture notes /Interview Q&A using Audio / Video Transcription

Students and job seekers often spend a significant amount of time listening to lengthy lectures or interview recordings. Manually writing notes or identifying important interview questions is time-consuming and may result in missing key information.

An intelligent transcription system can automatically:

- Convert speech into text.
- Extract important concepts.
- Generate lecture notes.
- Predict interview questions.
- Suggest suitable answers.

This saves time and improves productivity

Objective :-

The main objectives of this use case are:

- Convert audio/video into text.
- Generate concise lecture notes.
- Identify key topics discussed.
- Predict possible interview questions.
- Generate answers based on transcript content.
- Improve learning efficiency and interview preparation.

Step 1:- Input

The system accepts:

- Audio recordings (.mp3, .wav)

- Video recordings (.mp4, .avi)
- Transcript

Step 2: Speech-to-Text Conversion/ text- to - speech

The uploaded audio/video is processed using Speech Recognition models such as:

- OpenAI Whisper
- Google Speech-to-Text
- Azure Speech Service

The spoken content is converted into text.

Example :- Machine Learning is a subset of Artificial Intelligence that enables computers to learn from data without explicit programming.

Step 3: Text Preprocessing :- The transcript is cleaned by: - Removing filler words

- um
- uh
- okay
- you know
- Removing punctuation errors
- Sentence segmentation
- Tokenization
- Stop-word removal

The cleaned transcript becomes easier to analyze.

Step 4: NLP Processing

Natural Language Processing techniques identify:

- Keywords
- Important concepts
- Named entities
- Frequently discussed topics
- Context

Step 5: Lecture Note Generation

The system summarizes the transcript into structured notes.

Step 6: Interview Question Prediction

Using NLP and Large Language Models, the system predicts likely interview questions.

Example Questions:

- What is Machine Learning?
- Explain supervised learning.
- Difference between AI and ML.
- What is overfitting?
- Explain regression.

Step 7: Answer Generation

The system also predicts appropriate answers.

Step 8: system Architecture:-

**Audio vedio Input → speech to text → transcription generation →text preprocessing
→NLP →lecture notes generation →Interview ques prediction →Answer generation .**

Step 9: skills :-

Machine Learning

Upload audio or video files.

Automatic speech transcription.

Speaker identification (optional).

Lecture note generation.

Topic extraction.

Keyword extraction.

Summary generation.

Interview question prediction.

Automatic answer generation.

Export notes as PDF or Word.

Search within transcripts.

Step 10:- Future enhancement :-Real-time transcription during live lectures or interviews.

Multi-language transcription and translation.

AI-powered personalized study notes.

Automatic flashcard generation.

Voice emotion and sentiment analysis.

Integration with Learning Management Systems (LMS).

Cloud-based storage and synchronization.

AI chatbot for answering transcript-related questions.

Stake holders:-

Identify stake holders:

Directly affect-social media users / students

- Students experiencing stress due to excessive social media usage
- Mental health counselors
- School/College wellness teams
- Psychologists and therapists
-

Indirectly Affected-

- parents / Guardians
- Faculty members
- Department Heads
- School / College Administration
- Social Media Platform Providers
- Researchers studying digital well-being

Technical Stakeholders-

- IT Department
- Data Analysts
- AI/ML Development Team
- Data Engineers
- Database Administrators
- System Administrators
- Cybersecurity & Privacy Team

Business / Decision-Making Stakeholders –

- College Management
- Student Affairs Department

What ?- Problem cause:-

The need for predicting lecture notes and interview Q&A from audio/video transcription arises due to several common challenges:

1. Manual Note-Taking is Time-Consuming

- Students often struggle to write notes while simultaneously paying attention to lectures.
- Important concepts may be missed because of divided attention.

2. Lengthy Audio and Video Content

- Online lectures, webinars, and interview recordings can last from 30 minutes to several hours.
- Reviewing the entire recording repeatedly is inefficient.

3. Difficulty Identifying Key Information

- Not all spoken content is equally important.
Users may find it hard to distinguish key concepts, definitions, and important questions from general discussion.

4. Lack of Structured Learning Material

- Raw transcripts are usually long and unorganized.
- Without summarization, finding specific topics or revision points becomes difficult.

5. Interview Preparation Challenges

- Candidates often do not know which questions are likely to be asked based on interview discussions.
- Reviewing multiple interview recordings manually takes significant time and effort.

6. Human Errors in Manual Transcription

- Manually converting speech to text is slow and prone to spelling mistakes, missing information, and inconsistencies.

7. Language and Accent Variations

- Different accents, speaking speeds, and pronunciations make manual understanding more difficult, especially in multilingual environments.

8. Background Noise and Poor Audio Quality

- Noise, overlapping conversations, or unclear recordings reduce comprehension and make manual note-taking harder.

9. Difficulty in Quick Revision

- Students and professionals need concise notes for fast revision before exams or interviews, but long transcripts are difficult to review quickly.

10. Increased Workload

- Teachers, students, recruiters, and professionals spend considerable time organizing, summarizing, and reviewing recorded content, reducing overall productivity.

Why ?- why need to solve this problem ?

Impact of Not Having Automatic Lecture Notes & Interview Q&A Generation

For Students / Job Seekers:

- Time-consuming manual note-taking.
- Difficulty concentrating on lectures while writing notes.
- Missing important concepts and key points.
- Poor revision due to unorganized study material.
- Reduced learning efficiency and productivity.
- Difficulty preparing for interviews effectively.

For Colleges / Universities / Training Institutes:

- Reduced learning outcomes among students.
- Increased workload for faculty in preparing study materials.
- Difficulty managing and organizing recorded lectures.
- Lower student engagement in online and hybrid classes.
- Inconsistent quality of lecture notes.
- Limited accessibility for students who miss lectures or have hearing impairments.
- Reduced effectiveness of digital learning resources.

For Organization:

- More time required to review interview recordings.
- Difficulty identifying important interview questions and candidate responses.
- Increased manual effort in documentation and evaluation.
- Reduced efficiency in recruitment and training processes.
- Delayed decision-making due to lengthy interview analysis.

Who?

This system can be used by:

- Students for converting lectures into notes
- Teachers for preparing study materials
- Professionals for meeting transcripts
- Researchers for analyzing recorded content

Where ?

The AI Transcription System can be used in:

- Educational institutions
- Online learning platforms
- Corporate meetings
- Research organizations
- Content creation and media industries